

Module 07: Third Normal Form (3NF)

Third Normal Form (3NF) & Transitive Dependency

Universal 3NF Condition: Super Key LHS or Prime Attribute RHS & 3NF Synthesis

The Golden Rule of Third Normal Form (3NF):

Relation R is in 3NF iff for every non-trivial functional dependency $X \rightarrow Y$, at least ONE holds:

Condition 1: X is a SUPER KEY of R (X is Key Determinant)

OR Condition 2: Y is a PRIME ATTRIBUTE (Y is part of some Candidate Key)

Transitive Dependency Hazard

Chained Non-Key Determination: $\text{Key} \rightarrow X \rightarrow Y$

Example: Student(Roll, DeptCode, DeptHOD)

$\text{Roll} \rightarrow \text{DeptCode}$ and $\text{DeptCode} \rightarrow \text{DeptHOD}$

Neither DeptCode is Super Key NOR DeptHOD is Prime!

Why 3NF Prohibits This:

- If HOD changes, multiple student rows must update.
- If all students in a department leave, HOD info is lost!

Decomposition fix: S1(Roll, DeptCode) & S2(DeptCode, DeptHOD)

Lossless Join + Dependency Preservation guaranteed in 3NF!

3NF Synthesis Algorithm (Lossless + Preserving)

Bernstein's Proven Polynomial Procedure

Step 1: Compute Canonical Cover F_c

Simplify F into minimal canonical cover F_c .

Step 2: Create Sub-Relations

For each FD $X \rightarrow Y$ in F_c , make $R_i = X \cup Y$.

Step 3: Ensure Candidate Key Inclusion

If no R_i contains a Candidate Key of R, add $R_{key} = CK$.

Step 4: Eliminate Subsumed Relations

If $R_j \subseteq R_k$, discard R_j . Output final schema!

> **Course:** Database Management Systems (AKTU Code: **BCS501**)

****Topic Focus:**** Third Normal Form Criteria, Removal of Transitive Dependency, Universal 3NF Test Condition ($\$X\$$ is Super Key OR $\$Y\$$ is Prime Attribute), Bernstein's 3NF Synthesis Algorithm, Gateway Classes Solved Numericals.

1. Third Normal Form (3NF) Formal Definition

Ek relation schema $\$R\$$ **Third Normal Form (3NF)** me tab hota hai jab:

1. Wo relation already **2NF** me ho.
2. Us relation me **koi bhi Transitive Dependency na ho**.

1.1 The Universal 3NF Test Condition (AKTU Standard)

Relation $\$R\$$ ke har non-trivial functional dependency $\$X \rightarrow Y\$$ ke liye, neeche diye gaye do conditions me se **kam se kam ek condition** true honi chahiye:

$\$X\$$ is a Super Key of $\$R\$$

$\$Y\$$ is a Prime Attribute of $\$R\$$

1. $\$X\$$ is a Super Key of $\$R\$$ (Determinant super key hai)

$\$X\$$ is a Super Key of $\$R\$$

2. $\$Y\$$ is a Prime Attribute ($\$Y\$$ kisi bhi candidate key ka hissa hai)

[!TIP]

Agar kisi FD me LHS Super Key nahi hai, lekin uska RHS Prime Attribute hai, toh wo dependency ****3NF me fully ALLOWED hai****! Yeh 3NF ka sabse bada relaxer hai jo ise BCNF se alag banata hai!

2. Transitive Dependency Hazard

Transitive Dependency: Jab ek non-prime attribute kisi doosre non-prime attribute par depend karta hai through candidate key:

$XY \twoheadrightarrow Z$

$X \twoheadrightarrow Y$ and $Y \twoheadrightarrow Z$

$X \twoheadrightarrow Z$

$XY \twoheadrightarrow Z$

(Where XY is not a super key and Z is non-prime).

Practical Hazard: Redundancy, Update Anomaly, aur Deletion Anomaly paida hoti hai.

3. Bernstein's 3NF Synthesis Algorithm

Yeh algorithm guarantee karta hai ki relation 3NF me decompose ho jaye aur saath hi **Lossless Join** aur **Dependency Preservation** dono properties 100% preserve rahein:

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Algorithm: 3NF_Synthesis(R, F)

1. Find Canonical Cover F_c of F .
2. For each FD $(X \rightarrow Y)$ in F_c :

Create a new relation schema: $R_i = X \cup Y$;

3. If none of the generated relations R_i contains a Candidate Key of R :

Create an additional relation R_{key} containing any candidate key of R ;

4. Eliminate redundant relations:

If $R_j \subseteq R_k$, remove R_j ;

5. Return the resulting set of relations.

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4. Architectural Diagram

5. Gateway Classes Solved AKTU Examination Problem

Problem: Given relation $R(A, B, C, D)$ with FD set:

$$F = \{ AB$$

$$\rightarrow C, \text{quad } C$$

$$\rightarrow D, \text{quad } D$$

$$\rightarrow A \}$$

Find all Candidate Keys, determine the highest Normal Form of R , and decompose into 3NF if not already in 3NF.

Step 1: Find Candidate Keys

Attributes on RHS: $\{A, C, D\}$.

Attribute B never appears on RHS $\implies B$ must be in every key!

$(AB)^+ = \{A, B, C, D\} = R \implies \mathbf{AB}$ is a Candidate Key.

$(BC)^+ = \{B, C, D, A\} = R \implies \mathbf{BC}$ is a Candidate Key (via C

$$\rightarrow D$$

$$\rightarrow A).$$

$(BD)^+ = \{B, D, A, C\} = R \implies \mathbf{BD}$ is a Candidate Key (via D

$$\rightarrow A, \text{ then } AB$$

$$\rightarrow C).$$

Candidate Keys: $\{AB, BC, BD\}$.

Prime Attributes: $\{A, B, C, D\}$ (All attributes are Prime!).

Non-Prime Attributes: $\emptyset$ (None!).

Step 2: Determine Highest Normal Form

Let us test each FD in F :

1. AB

$$\rightarrow C:$$

- Is LHS AB a Super Key? **YES** (AB is a Candidate Key). $\implies$ Satisfies 3NF and BCNF.

2. C

$$\rightarrow D:$$

- Is LHS C a Super Key? **NO** ($C^+ = \{A, C, D\}$

$\neq R$).

- Is RHS D a Prime Attribute? **YES** (D is part of Candidate Key BD). $\implies$ **Satisfies 3NF!**

3. D

$$\rightarrow A:$$

- Is LHS D a Super Key? **NO** ($D^+ = \{A, D\}$

$\neq R$).

- Is RHS A a Prime Attribute? **YES** (A is part of Candidate Key AB). $A \rightarrow B$ implies **Satisfies 3NF!**

Conclusion:

Relation R is in **3NF** because for every FD where LHS is not a super key, RHS is a prime attribute!

Relation R is **NOT in BCNF** because in C

$A \rightarrow D$ and $D \rightarrow A$

$A \rightarrow C$, the determinants C and D are not super keys!

Highest Normal Form = 3NF.